

Sporamin induce apoptosis in human tongue carcinoma cells by down-regulating Akt/GSK-3 signaling.

We investigated the effects of sporamin, the major soluble protein with a kunitz-type trypsin inhibitory activity in the root tuber of the sweet potato, on cell proliferation, apoptosis, Akt/GSK-3 signaling and its related genes to provide more insights in the mechanism behind the inhibitory effects of sporamin in a human tongue cancer line Tca8113. In this study, sporamin inhibited cell proliferation and induced apoptosis in Tca8113 cells in a concentration-dependent and time-dependent manner. Consistently, Bax was up-regulated and Bcl-2 was down-regulated in sporamin-treated cells. Furthermore, Akt/GSK-3 signaling was down-regulated in sporamin-treated cells. Consistently, the phosphorylated Bad was significantly declined in sporamin-treated Tca8113 cells. These results suggest the antiproliferative effects of sporamin in Tca8113 cells might result partly from induction of apoptosis by down-regulating Akt/GSK-3 pathway. © 2010 The Authors Fundamental and Clinical Pharmacology © 2010 Société Française de Pharmacologie et de Thérapeutique.